Data Flow Diagrams (DFD)

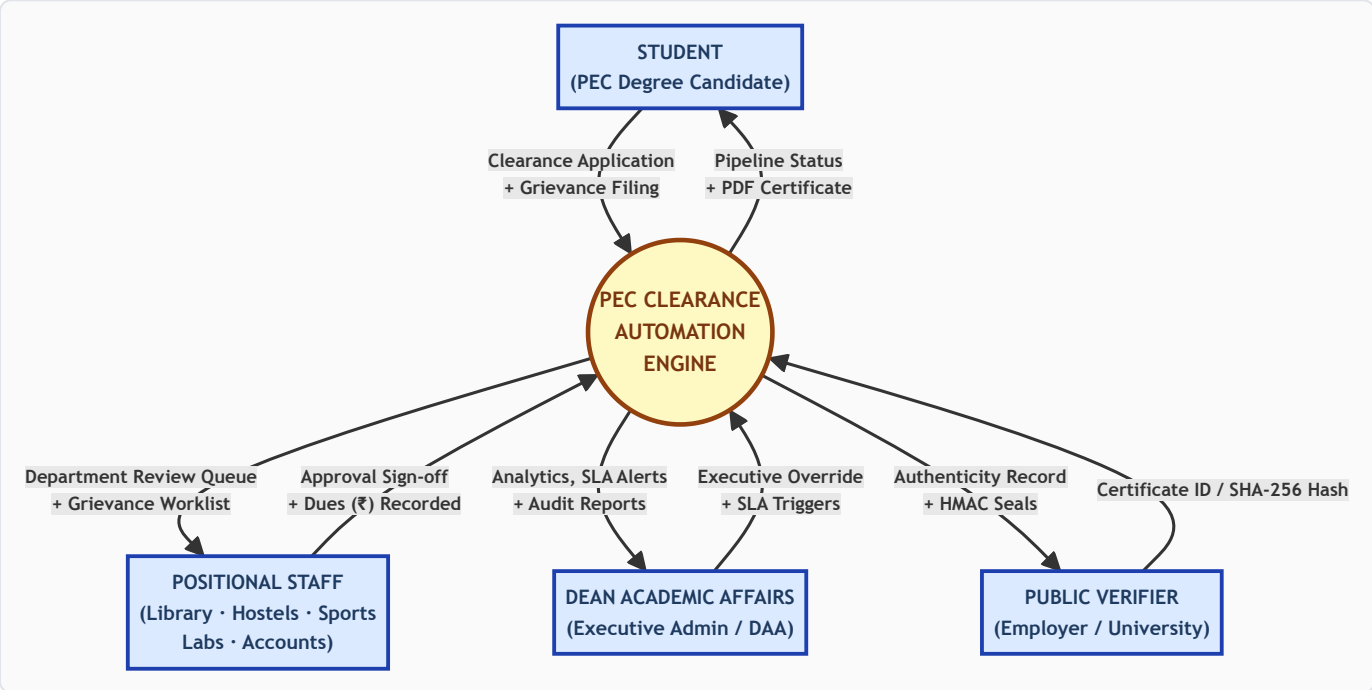
Academic Grievance, No-Dues & Clearance Automation Engine

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Course: Software Engineering
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Notation: Yourdon / De Marco — Circles = Processes, Rectangles = External Entities, Open Boxes = Data Stores

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1. DFD Level 0 — Context Diagram

The Level 0 Context Diagram treats the entire system as a **single process bubble** and maps every interaction between the system and the four external entities on campus.



Level 0 Explanation

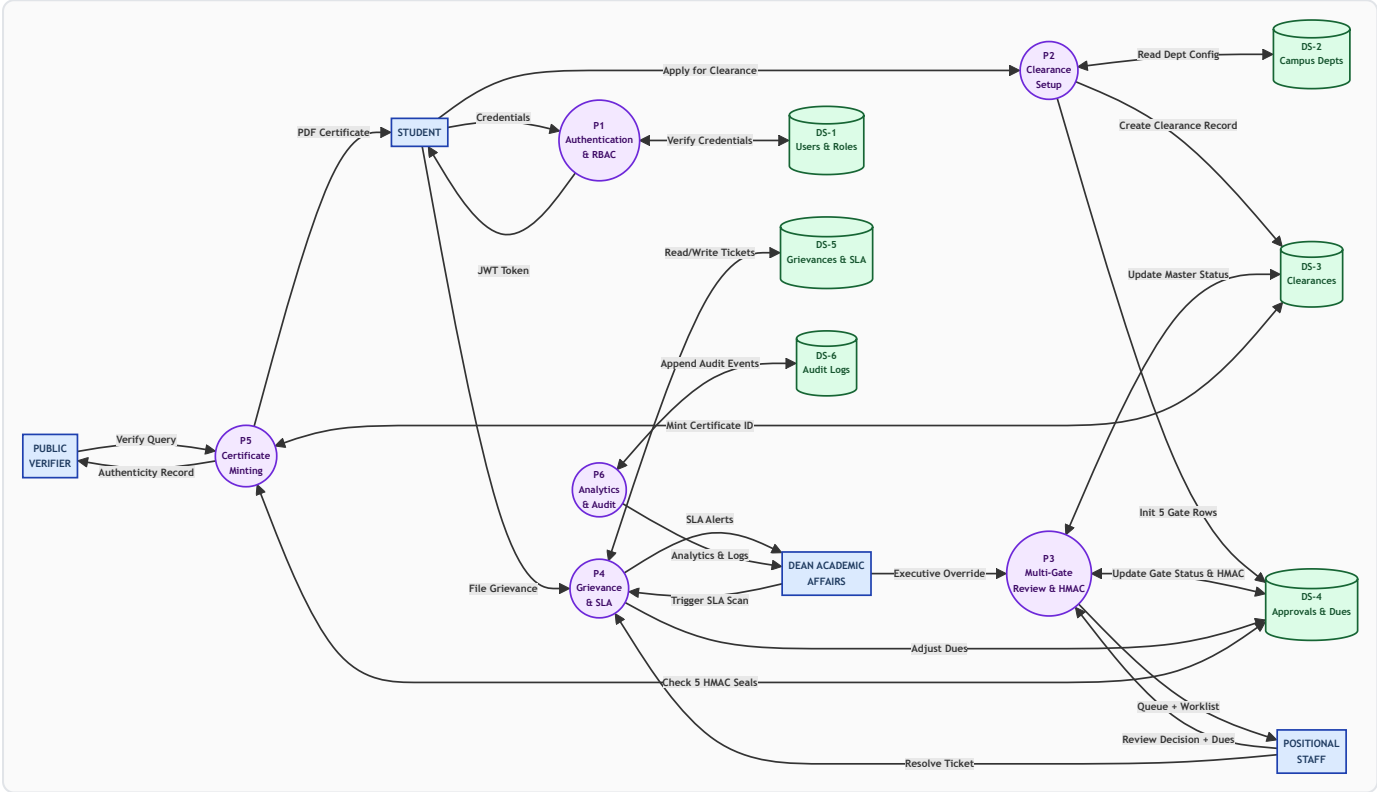
External Entity	Role	Data Flows In	Data Flows Out
Student	Degree candidate applying for clearance	Clearance application, grievance filings	Live pipeline status, official PDF certificate
Positional Staff	Departmental signatories (Library, Hostels, Sports, Labs, Accounts)	Approval/rejection decisions with dues in ₹	Review queue, grievance worklist
Dean Academic Affairs	Executive override authority, monitors SLA	Override orders, SLA trigger commands	Campus analytics, SLA breach alerts, audit logs

External Entity	Role	Data Flows In	Data Flows Out
Public Verifier	Employer or university checking certificate	Certificate ID or SHA-256 hash	Authenticity JSON with HMAC seals

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2. DFD Level 1 — System Decomposition

The Level 1 DFD decomposes the single engine bubble into **six functional processes** connected by data flows and six persistent data stores.



Process Descriptions

Process	Name	Core Function
P1	Authentication & RBAC	Validates credentials against Bcrypt hashes in DS-1; issues signed JWT with role claims (STUDENT / STAFF / ADMIN).
P2	Clearance Setup	Enforces single-active-request rule; creates master clearance record in DS-3; initialises 5 gate approval rows in DS-4 (Gate 1 → PENDING, Gates 2 & 3 → WAITING).
P3	Multi-Gate Review & HMAC	Orchestrates Chain of Responsibility pipeline; enforces sequential gate barriers; computes HMAC-SHA256 seals on approval; handles DAA overrides.
P4	Grievance & SLA Engine	Accepts student dispute filings; assigns dynamic SLA deadlines (CRITICAL 4 h, HIGH 12 h, MEDIUM 24 h, LOW 48 h); auto-escalates breached tickets to DAA.
P5	Certificate Minting	Verifies all 5 HMAC seals; unlocks ₹10,000 Caution Money refund; mints unique Certificate ID; generates SHA-256 public digest; produces official PDF.

Process	Name	Core Function
P6	Analytics & Audit	Records every security and operational event to the append-only DS-6 audit log; computes campus-wide clearance metrics.

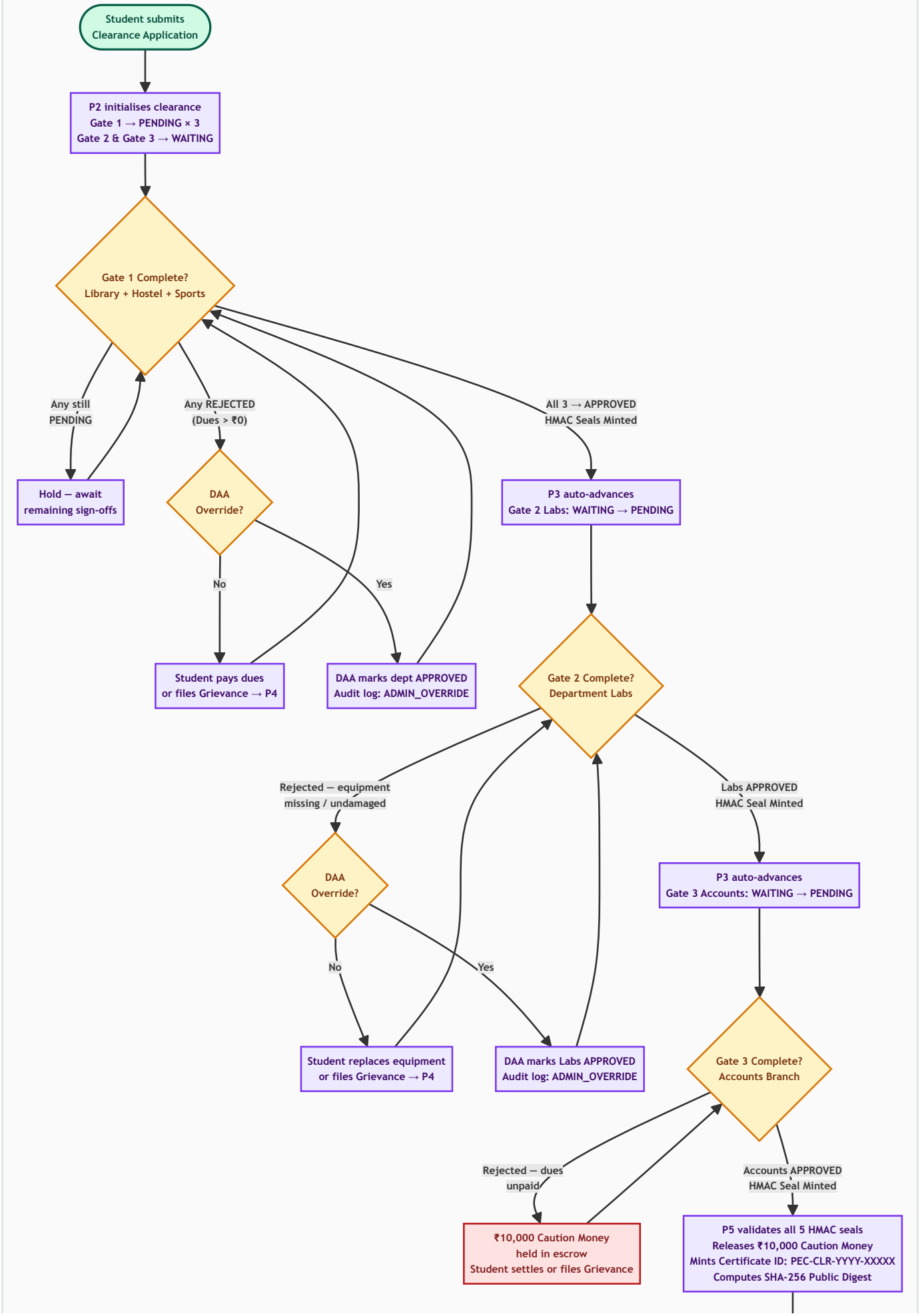
Data Store Descriptions

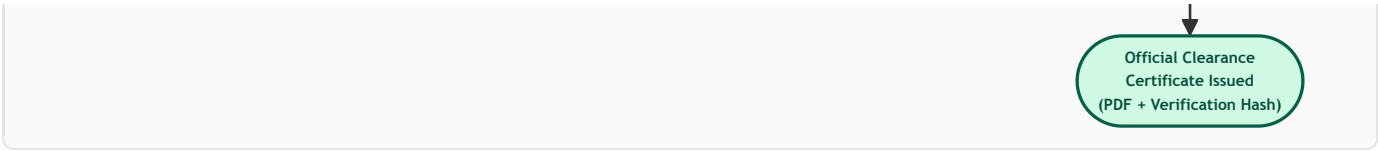
Store	Table	Contents
DS-1	`users`	Credentials, Bcrypt hashes, role, SID, department assignment.
DS-2	`departments`	5 facility records — gate classification, head title, department code.
DS-3	`clearance_requests`	Master clearance record — student ref, status, certificate ID, SHA-256 digest.
DS-4	`department_approvals`	Per-gate rows — status (WAITING / PENDING / APPROVED / REJECTED), due amount (₹), HMAC seal.
DS-5	`grievances`	Dispute tickets — priority, SLA timestamp, escalation flag, staff response.
DS-6	`audit_logs`	Immutable append-only log — actor, action verb, entity, UTC microsecond timestamp.

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3. Core Decision Tree — Multi-Gate Clearance Pipeline

The Decision Tree traces the **deterministic gate-by-gate path** from a student's initial application to the final Clearance Certificate, capturing every branch condition enforced by the Chain of Responsibility engine.


Student submits Clearance Application
P2 initialises clearance Gate 1 → PENDING × 3 Gate 2 & Gate 3 → WAITING
Gate 1 Complete? Library + Hostel + Sports
Any still PENDING
Hold — await remaining sign-offs
Any REJECTED (Dues > ₹0)
DAA Override?
No
Student pays dues or files Grievance → P4
Yes
DAA marks dept APPROVED Audit log: ADMIN_OVERRIDE
All 3 → APPROVED HMAC Seals Minted
P3 auto-advances Gate 2 Labs: WAITING → PENDING
Gate 2 Complete? Department Labs
Rejected — equipment missing / undamaged
DAA Override?
No
Student replaces equipment or files Grievance → P4
Yes
DAA marks Labs APPROVED Audit log: ADMIN_OVERRIDE
Labs APPROVED HMAC Seal Minted
P3 auto-advances Gate 3 Accounts: WAITING → PENDING
Gate 3 Complete? Accounts Branch
Rejected — dues unpaid
₹10,000 Caution Money held in escrow Student settles or files Grievance
Accounts APPROVED HMAC Seal Minted
P5 validates all 5 HMAC seals Releases ₹10,000 Caution Money Mints Certificate ID: PEC-CLR-YYYY-XXXXX Computes SHA-256 Public Digest



Decision Tree Explanation

The three-gate pipeline enforces **strict sequential dependency barriers**:

- **Gate 1 — Parallel Departmental Stage:** Library, Hostels, and Sports departments operate concurrently. Any rejection blocks further pipeline advancement and routes the student to either pay dues or file a formal grievance (handled by Process 4). The Dean Academic Affairs (DAA) holds an executive override authority at every gate for cases of administrative error or approved waivers, with a mandatory audit trail entry recorded in DS-6.

- **Gate 2 — Academic Labs Stage:** Automatically unlocked only when all three Gate 1 facilities are `APPROVED`. The HOD of Department Labs verifies equipment returns; approval seals the record with an HMAC-SHA256 signature. Any rejection loops back unless overridden by the DAA.

- **Gate 3 — Accounts & Certification Stage:** Automatically unlocked only after Gate 2 is `APPROVED`. The Accounts Branch reconciles all outstanding institutional dues. Upon approval, the ₹10,000 Caution Money deposit is released, Process 5 validates all five HMAC seals, and the system generates the unique Certificate ID and public SHA-256 verification digest — producing the official, tamper-evident PDF clearance certificate.